

Investigation on AEB Key Parameters for Improving Car to Two-Wheeler Collision Safety Using In-Depth Traffic Accident Data

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Abstract—Automatic emergency braking (AEB) system play an essential role in avoiding traffic collisions and minimizing impact strengths, which contributes to improving road traffic safety. However, the present research of the AEB system is mainly concentrated on avoiding the car to car and car to pedestrian collisions. China has typical mixed traffic characteristics, especially the frequent car to two-wheeler collision accidents. Research on the AEB system to prevent car to two-wheeler collisions in complex multi-traffic scenarios is of great significance to further improve the comprehensive performance of the AEB system. In this article, based on the cases of the car to two-wheeler collisions from China In-Depth Accident Study (CIDAS) database, the effects of three key parameters of the AEB system, braking deceleration, braking advance time, and detection range, on the accident rate and severity are studied by accident reconstruction and virtual experiment. Results show that the theoretical optimal detection range of the AEB system is 180°–35 m for the car to two-wheeler collision prevention. Moreover, the current feasible optimal detection range is 120°–35 m due to the limitation of environmental awareness technology. When the braking advance time is 1 s, the detection angle is 120°, and the detection distance is 35 m, the installation of the AEB system can avoid 22.3%

of the car to two-wheeler collision accidents. Meanwhile, in the inevitable accidents, the average collision speed of vehicles decreases from 33.93 to 21.34 km/h, which effectively reduces the collision strengths. Findings of this study could help car manufacturers and makers of AEB hardware select the appropriate detection radar. They can also provide ideas and data support for formulating relevant laws and regulations of the AEB system, future development of AEB system, or other driver assistance systems.

Index Terms—Intelligent-assisted driving, automatic emergency braking (AEB), detection range, braking advance time, car to two-wheeler collision.

I. INTRODUCTION

THE increase in urbanization has brought about a corresponding increase in urban road construction. Considering the problem of road congestion, an increasing number of people choose two-wheelers as their preferred travel tools. Statistical data published by the Roads and Traffic Authority of the Ministry of Public Security show that China has one of the highest rates of accidents involving two-wheelers. There were 45,868 motorcycle road traffic accidents in 2018 that caused 55,071 injuries and 10,663 deaths. Additionally, the data show 1,840 bicycle road traffic accidents that caused 1,720 injuries and 372 deaths [1]. As people emphasize more on travel safety, the active safety of vehicles has become a popular topic for scholarly research [2]. The protection of motorcyclists and cyclists is also a key issue in the study of vehicle active safety [3], [4].

Automatic emergency braking (AEB) system is preventive security system. When a vehicle is at risk of colliding with obstacles, such as a two-wheelers, pedestrians, or cars, AEB system assists the driver in taking action [5], [6]. Since 2014, a new car evaluation agency has added AEB system to its latest car rating system applied at major new vehicle evaluation institutions [7], [8], [9], [10]. In 2016, the evaluation of AEB system on pedestrian safety was adopted [11]. The European New Car Assessment Program (E-NCAP) launched the AEB test evaluation project for bicycle targets in the 2018 version of E-NCAP, mainly focusing on two-wheeler collision assessment for AEB system [12]. New car assessment procedures have been actively studied and established in domestic medium and long-term development projects. Due to the frequency of two-wheeler accidents, new car assessment procedures are paying increased attention to vulnerable road users. The study of AEB system that prevents two-wheeler accidents is an essential part of this process.

AEB system relies on sensors to detect road conditions and make corresponding driving decisions, and their performance

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significantly depends on the reliability of these sensors [13]. Many scholars have been studying various cases of real traffic accidents and the corresponding depth data of AEB system [14], [15], [16], [17], [18]. Indian scholars have conducted benefit evaluations of AEB system on cars under different early warning times, triggering mechanisms and speed reductions based on the Road Accident Sampling System–India (RASSI) database [19]. The effectiveness of the AEB system depends on its radar's function [19]. British scholars analyzed collisions between cars and two categories of vulnerable road users (pedestrians and two-wheeler riders) based on a project in the British Ministry of Transport and Highway Administration to understand road traffic accidents, and the optimal value of the system's detection range can be determined according to its relative position to other vehicles 1, 2, and 3 s before collision [20]. Zhao et al. collected traffic recorder data and used PC-Crash to reconstruct 40 cases of vehicle to two-wheeler collision accidents at intersections, they then changed the field of view of the sensor and delay time of starting vehicle deceleration and reconstructed these accidents again to obtain the detection range of the system [21]. Feng et al. investigated the detection range of AEB system based on data from the SHUFO joint road traffic safety science research center in Shanghai for rear-end collision accidents, and they improved the AEB system by optimizing its parameter ranges [22].

Significant differences are found in the traffic between China and foreign countries that China has typical mixed traffic characteristics [23], [24], [25], [26], [27]. Additionally, China is one of the most accident-observed countries involving two-wheeler, the safety issue of two-wheeler riders needs to be addressed urgently [1]. The AEB system can effectively avoid road accidents and reduce the severity of accidents to a certain extent. However, existing studies on the AEB system mainly focused on the avoidance of car to car and car to pedestrian collisions, while how to avoid two-wheeler-related accidents needs more attention [22], [28], [29], [30], [31]. Additionally, in the AEB system study, many scholars only considered the relative position or time to collision (TTC) [19], [20], [21], which are different from the actual working process of the AEB system. They also ignored the influence of different driving behavior that may lead to poor reliability and accuracy of the final results.

Given the above background and scientific issues, this study screened out cases of car to two-wheeler collision from China In-Depth Accident Study (CIDAS) database. These two-wheelers include bicycles, electric two-wheelers, and motorcycles, which are in line with the complex mix of vehicles on the road in China. Our main contributions are listed in the followings.

- In view of the existing research less attention AEB to prevent car to two-wheeler collision, this study proposes to explore the optimal value of AEB key parameters in complex traffic scenarios based on in-depth traffic accident data.
- To be closer to the actual AEB implementation process and get reliability and accuracy results, this study innovatively proposed to first study the two key parameters of brake deceleration and brake advance time, and then combine these two key parameters with the relative position to jointly analyze the detection range of the AEB system.

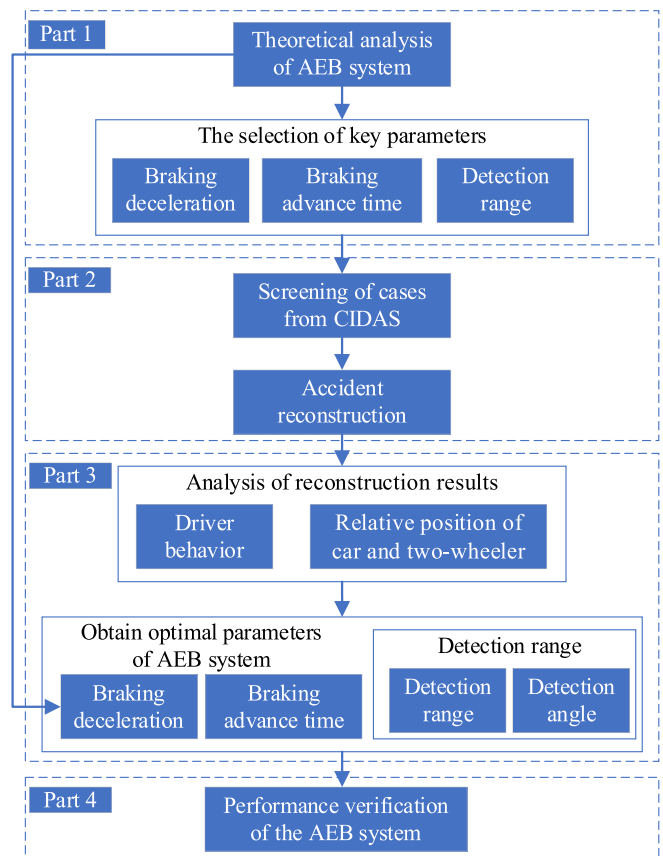


Fig. 1. Block diagram of the overall structure.

- Additionally, this study is the first to propose that drivers' driving behavior should be considered when studying the key parameters of the AEB system.

Fig. 1 shows the block diagram of the overall structure. The remainder of the paper is organized as follows: (1) Theoretical analysis of the AEB system to determine the key parameters for the car to two-wheeler collision safety; (2) Screening of the car to two-wheeler collision accident cases from CIDAS database and accident reconstruction of these cases using PC-Crash software; (3) Analysis of reconstruction results to obtain optimal parameters of the AEB system; (4) The AEB system with optimal parameters are applied to that car to two-wheeler collision accident cases, and the improvement in the AEB system's performance is verified.

II. DATA AND METHODS

A. AEB System and Parameters Selection

The AEB system is based on the forward collision warning (FCW) system and works closely with the braking system [32], [33]. Fig. 2 shows its working principle. The FCW system uses radar to monitor the vehicle in front of it and judges the distance, azimuth, and relative speed between the vehicle in front of it. It provides warning information when there is a potential collision danger, and the braking system acts as an emergency brake when warning information is received [33].

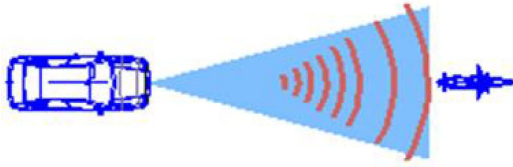


Fig. 2. Working diagram of the AEB system.

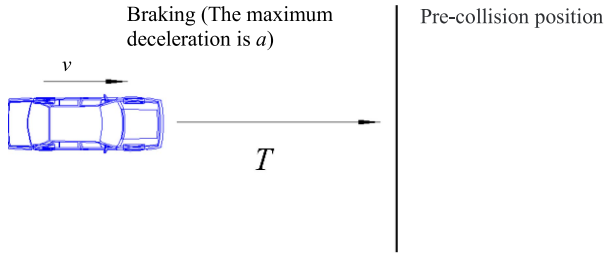


Fig. 3. Diagram of the braking process under the intervention of the AEB system.

The vehicle collision speed is closely related to the degree of injury in collisions [34]. The greater the collision speed is, the more severe the injuries are [35]. Early and late braking lead times severely affect the collision speed and inevitably affect the protection level of vulnerable road users. Thus, the protection effect is directly related to the performance of the AEB system.

When the TTC of the vehicle is equal to the braking advance time (T), the AEB system intervenes to brake the vehicle automatically and avoid collisions. However, if the detection range of the system is not sufficient to detect the target object, the system cannot determine the presence of dangerous conditions, leading to system underperformance. Therefore, this study investigates the effect of the detection range of the system combined with the braking advance time to ensure the protection effect of the AEB system for vulnerable road users.

The two extremely important functions of the AEB system are perception and braking. How to quickly and accurately identify all potentially dangerous pedestrians and vehicles ahead in the complex traffic environment and stop the car in time are the most important tasks of the AEB system. To achieve this goal, determining the optimal detection range of the FCW detection radar is necessary under the premise of knowing the braking advance time and deceleration of the braking system. Therefore, the braking advance time, braking deceleration, and detection range of the AEB system are key parameters related to the comprehensive performance of the AEB system.

Fig. 3 shows a schematic of the braking process under the intervention of the AEB system. The pre-collision position in the figure is the location where the collision occurs. Suppose that the AEB system starts to brake the vehicle at the time when the vehicle's speed is V , the maximum braking deceleration is a , and the braking advance time is t . The collision is avoided if the vehicle can reach a speed of 0 before the pre-collision position. It means that the collision is avoided when the TTC of the vehicle is greater than or equal to the braking advance time

(T). The collision time is given as follows:

$$TTC = \frac{s}{v} \quad (1)$$

In (1), s represents the distance from the vehicle to the pre-collision position (in m); v represents the vehicle speed (in m/s).

In [30], the braking process was divided into the driver response stage, the braking coordination stage and the continuous braking stage. The present study does not consider whether the driver reacts when the collision is about to occur. Therefore, we consider only the braking coordination and the continuous braking stages for the braking process.

$$T = t_1 + t_2 \quad (2)$$

In (2), t_1 represents the brake coordination time, in s; t_2 represents the brake duration, in s. First, the braking coordination time t_1 is determined to obtain the braking advance time of the car. According to Yang et al. [30], we set the brake coordination time to 0.3 s in this study, i.e., $t_1 = 0.3$ s. Then, t_2 is determined as follows:

$$t_2 = -\frac{v}{a} \quad (3)$$

The vehicle speed, v , can be obtained through the CIDAS data or accident reconstruction. The braking deceleration, a , is affected by the road conditions and cannot be determined directly.

In the actual driving process, most roads that automobiles use are asphalt or cement pavements. The braking adhesion coefficient of automobiles on these road ranges from 0.55 to 0.7. Therefore, the maximum range of the braking deceleration that can be achieved is 5.4–6.9 m/s^2 [36]. On snowy days or slippery roads, the brake adhesion coefficient of the car does not necessarily reach 0.55. According to our previous research results, in domestic collision accident scenarios, fewer accidents between cars and two-wheelers are found on snowy days or slippery roads, possibly because two-wheelers are used less under these conditions. Therefore, this study considers only braking deceleration under ideal conditions (such as a dry surface or efficient braking system). Considering the limitations on the safety and comfort of passengers and common realistic conditions, we use a braking deceleration value of 6 m/s^2 .

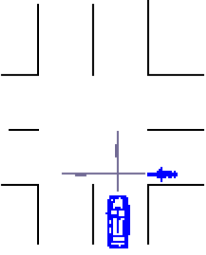
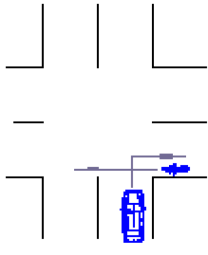
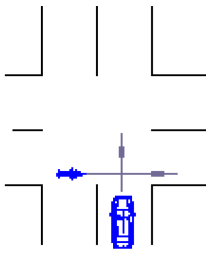
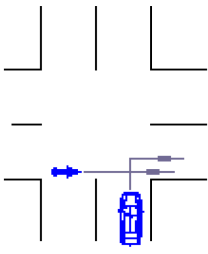
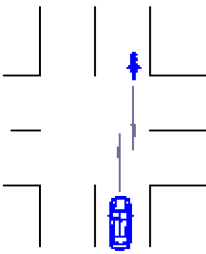
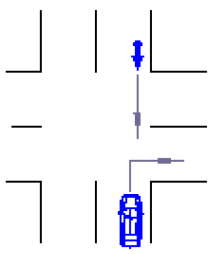
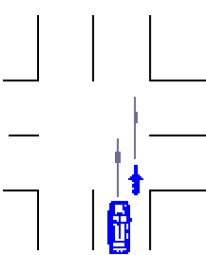
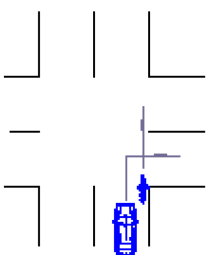
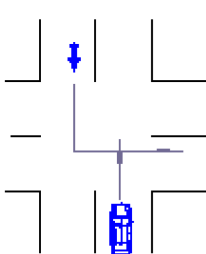
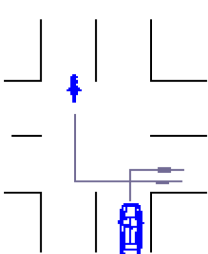
B. Accidents Data for AEB Key Parameters Analysis

China has typical mixed traffic characteristics, especially the frequent car to two-wheeler collision accidents. In this study, we use the in-depth traffic accident data to determine AEB key parameters to improve car to two-wheeler collision safety.

The China Automobile Technology Research Center's "China in-depth Accident Study (CIDAS)" project was initiated by many well-known automobile enterprises inside and outside the United Nations. In this paper, we collected the car to two-wheeler collision accidents from 2011 to 2020 from the CIDAS database as the research object.

Our previous research results [24] show that in most car to two-wheeler collision accidents, the two-wheeler is in front of the car when the accident occurs [36]. Table I shows the

TABLE I
SCHEMATIC DIAGRAM OF ACCIDENT SCENARIOS

	Schematic diagram of the accident scene	Remarks	The number of cases		Schematic diagram of the accident scene	Remarks	The number of cases
1		Occurs at intersections or non-intersections	49	6		Occurs at any intersection	12
2		Occurs at intersections or non-intersections	47	7		Occurs at any intersection	11
3		Occurs at intersections or non-intersections	16	8		Occurs at any intersection	1
4		Occurs at intersections or non-intersections	20	9		Occurs at any intersection	14
5		Occurs at intersections or non-intersections	12	10		Occurs at any intersection	44

summarized schematic of these accident scenarios. Therefore, the type of collision shown in Fig. 4 is one of the criteria for case selection in this study. The motion state of the two-wheeler is more complex; within 2 s of a collision, a two-wheeler may move to the rear of the vehicle or may be blocked by other obstacles [24].

In summary, considering the actual traffic environment in China, the following criteria were used for case selection and selected 301 accident cases.

- 1) Accidents occurred from September 2011 to June 2020.
- 2) The involved vehicles included only one car and one two-wheeler.

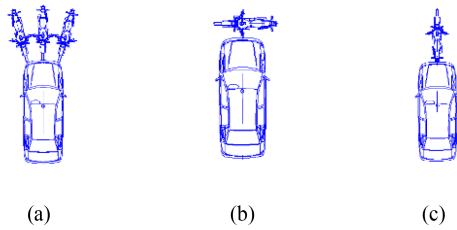


Fig. 4. Schematic of the collision types. (a) The front of a car collides with the front of a two-wheeler. (b) The front of a car collides with the side of a two-wheeler. (c) The front of a car collides with the rear of a two-wheeler.

- 3) The two-wheeler types included bicycles, electric two-wheelers, and motorcycles.
- 4) The other vehicles included cars, vans, and SUVs.
- 5) The accident collision types are as shown in Fig. 2.
- 6) The accidents did not include two-wheeler colliding with the rear of the car or two-wheeler obscured by obstacles within 2 s before the collision.

According to the people involved in the accidents, 92 of these 301 accidents were severely injured, accounting for 30.6% and 19 fatal accidents, accounting for 6.3%. There were only 190 minor accidents, accounting for 63.1%.

C. Reconstruction Methods

PC-Crash is an accident reconstruction software developed by the Austrian DSD company based on the basic theories of dynamics and kinematics, such as momentum and impulse conservation. It is currently one of the most widely used software in the field of traffic accident reproduction [37]. The software can simulate multiple types of accidents, including two-dimensional and three-dimensional collisions. It can input vehicle speed, vehicle position, braking delay time, and the driver's control of the vehicle's brake pedal and accelerator pedal. It can also output vehicle trajectory and impact. Speed, stopping, and position have the characteristics of short calculation time, high modeling efficiency, and high reliability of modeling results.

Based on the accident information obtained from the CIDAS database, 301 cases of car to two-wheeler collision accident were reconstructed using the PC-crash traffic accident reconstruction software. This is to obtain driver's behavior and the relative position of two rounds of a car within 2 s before the collision, and combined with the analysis of the content of the above parameters, analyze the effects of AEB system key parameters of braking advance time and detection range to accident severity and accident rate, and obtain the theoretical and technical optimization values of the detection range.

For this purpose, the following general steps were followed:

- 1) *Import of Accident Environment Information*: This included importing the accident map, adjusting its scale, setting the friction coefficient, loading the vehicles, modifying parameters, and tire selection, etc.
- 2) *Post-Collision Simulation*: This pertained to the definition of the event sequence after the collision, the setting of collision parameters, and simulation; the latter involved

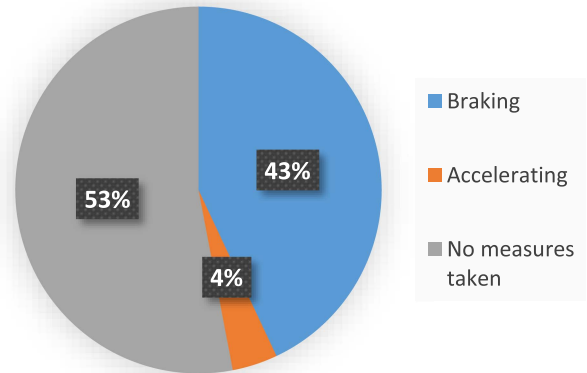


Fig. 5. Driver behavior statistics at 2 s before a collision.

constantly modifying the parameters until the simulation results were basically consistent with the site information.

3) *Pre-Collision Simulation*: This included setting up the simulation model, kinematics simulation, pre-collision sequence definition, collision parameter setting, and simulation starting 4.7 s before the collision, as per CIDAS requirements.

This section focuses on expounding the preliminary theoretical and data preparation work of this research, as well as the mid-term simulation experiment process, which lays the foundation for the output of subsequent results and conclusions. First, we introduced the working diagram of AEB system and explained why these parameters are key parameters. And then we expounded the meaning, calculation method of these key parameters and the basis for determining the values of some parameters. Next, we introduced the source, the screening standard of accident data, the accident type and severity in these data. At last, we introduced the reconstruction methods using PC-Crash software.

III. RESULTS

A. Reconstruction Results

1) *Driver Behavior*: After performing the above operations, we obtained the collision speed of the vehicles involved in the accident at the moment of collision and the motion response and motion process of the vehicles before and after collision [38], [39], [40].

The driver's behavior at 2 s before the collision was elicited based on accident reconstruction results. Fig. 5 shows the statistical results. Among the 301 car to two-wheeler collision accidents, 129 drivers took braking measures within 2 s before the collision, accounting for 43% of the accidents. This shows that in 43% of accidents the drivers perceived danger and started braking, but accidents and casualties occurred nonetheless. In 11 accidents, the drivers accelerated, which may have been a wrong response by nervous drivers or drivers trying to avoid the accident through acceleration. In the remaining 161 accidents, the drivers did not take any measures; these accounted for 53% of the accidents. In addition to the drivers' response to

dangerous conditions, these data further illustrate that corresponding automatic measures should be applied systematically to avoid accidents.

2) *Relative Position of Car and Two-Wheeler*: We extracted the absolute coordinate data of the car and two-wheeler from the PC-Crash software. Then, we performed a coordinate transformation via Excel software to obtain the relative position data of the car and two-wheeler. Furthermore, we used Origin software to draw the graphics.

The relative positions of the car and two-wheeler were at 1, 1.5, and 2 s before the collision and marked individually, as shown in Fig. 6(a)–(c), respectively. The red points in the figure indicate the position of the two-wheeler relative to the vehicle in accidents that resulted in the drivers' death. By contrast, the black points correspond to accidents that resulted in injuries.

B. Braking Advance Time

We obtained the braking advance time of each accident using (2). Fig. 7 shows the braking advance time cumulative percentages. As shown in the figure, the black curve is based on the braking advance time of each accident. It was calculated for each percentile after summation for all cases. The red curve is the fitting line of the black curve.

Fig. 6 shows that without considering the influence of other factors, 44% of accidents can be avoided if the braking advance time is 1 s. Meanwhile, 83% and 96% of the accidents can be avoided if the braking advance time is 1.5 s and 2 s, respectively. However, too much braking advance time may interfere with the normal driving of the car and vehicles around it, leading to more danger. Therefore, although about 96% of the accidents can be avoided when the braking lead time is 2 s, a braking lead time of 2 s is too long for accidents that require less than 1 s to be avoided. This may lead to driver dissatisfaction with the AEB system or other accidents. Therefore, this study only considers braking advance times of 1, 1.5, and 2 s. In our previous study, we considered a time of only 2 s before the collision because the maximum value of the set braking advance time was set to 2 s.

C. Detection Range

The AEB hazard detection function is closely related to the detection range of the sensor. Only when a dangerous obstacle appears in the detection range of the sensor can the detection function of the AEB system be realized so that its operation can proceed to the next stage. Because the detection area of general detection sensors is usually sector-shaped, the detection range and angle are commonly used to describe the detection area. Therefore, we investigate the detection range of the system from two aspects: detection distance and angle.

To prevent accident through the intervention of an AEB system, the braking advance time must be at least equal to TTC before the collision. When the braking advance time is 1 s, the TTC is 1 s. Therefore, this study considers the relative positions of the car and two-wheeler at 1, 1.5, and 2 s before the collision to explore the detection range of the AEB system, as shown in Fig. 6.

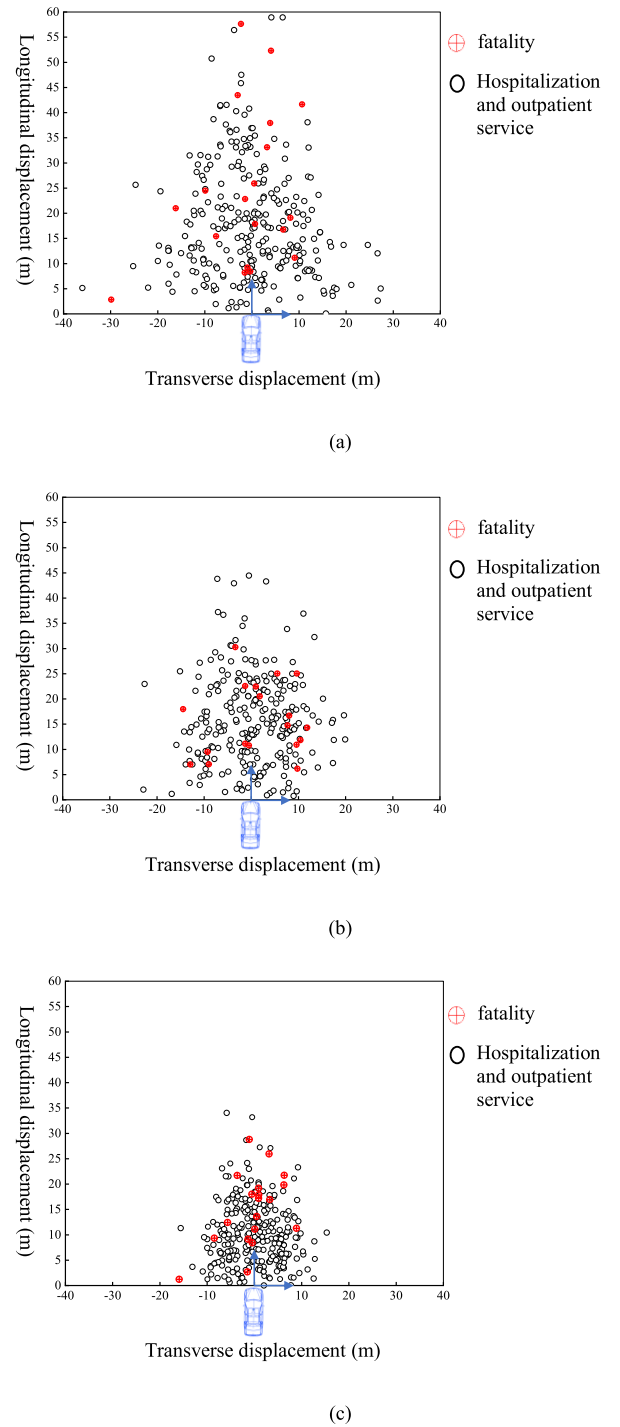


Fig. 6. Relative positions of the car and two-wheeler before the collision. (a) 2 s before the collision. (b) 1.5 s before the collision. (c) 1 s before the collision.

1) *Detection Angle*: Suppose that the cars are equipped with sensors. To study the detection angle of the AEB system for car to two-wheeler collision accidents, the visual contours of detection angles of 40°, 50°, 60°, 70°, 80°, 90°, 100°, 110°, 120°, 130°, 140°, 150°, 160°, and 170° were simulated on the relative position map of the car and two-wheeler of Fig. 3, as shown in Fig. 8. The probabilities of two-wheelers being detected at

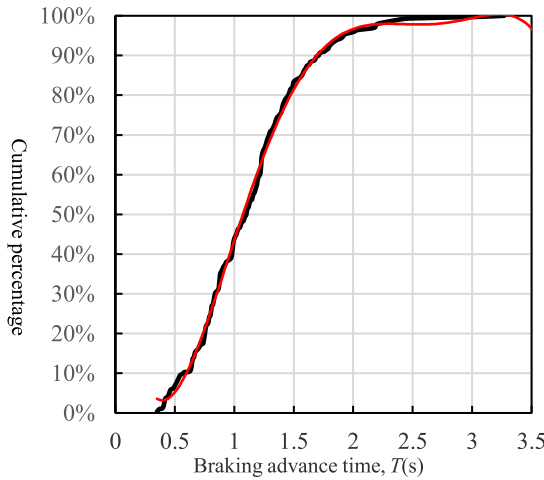


Fig. 7. Statistical chart of the accumulated percentage of brake advance time. Black curve: Cumulative percentage of brake advance time for all cases. Red curve: the fitting line of the black curve.

different detection angles were obtained using Fig. 8, as shown in Fig. 9.

As shown in Fig. 9, when the detection angle is 40° , the probability of detecting two-wheelers in fatal accidents does not increase with the decreasing TTC but first decreases and then increases. In accidents involving injuries, the two-wheeler at 1.5 s before the collision had the highest detection probability among the three-time points in almost all detection angles, reflecting the flexibility of the two-wheeler on the road and the complexity of the driver's driving behavior [47]. Consequently, the detection angle must reach 180° to detect all two-wheelers in these accidents. This is very different from car to pedestrian collision accidents, according to our previous results on car to pedestrian collision accidents, where the detection angle only needs to reach 40° to detect all pedestrians [24]. To ensure the effectiveness of the AEB system in protecting vulnerable road users, in theory, a detection angle of 180° is more appropriate. However, the larger the detection angle is, the lower the detection accuracy. Additionally, because of the increase in the number of target objects, the complexity of the control algorithm will increase, as will the hardware requirements. Therefore, we need to choose an optimal detection angle that will ensure the performance of the AEB system and reduce the equipment and computational complexity.

In extreme weather conditions, the penetration ability of LIDAR is poor, and millimeter-wave radars can make up for this shortcoming. Table II presents the parameters of some commercially available millimeter-wave radars. Current state-of-the-art millimeter-wave radars have a detection angle of about 120° , and this value should form the corresponding maximum for our study.

2) *Detection Distance*: Similarly, the accident cars are assumed to have been equipped with sensors. The visual contours for the detection distance of 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, and 60 m were simulated based on the relative position map Fig. 4, as shown in Fig. 10. Consequently, the probabilities of

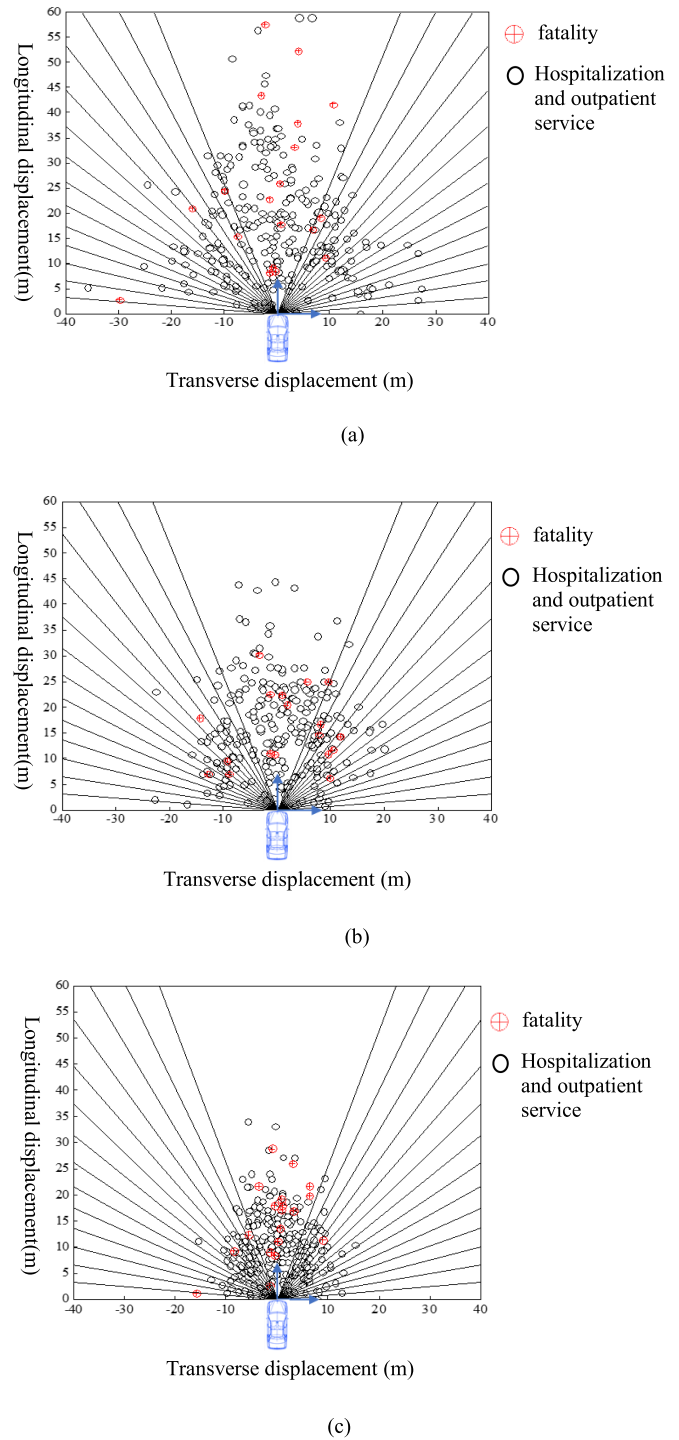


Fig. 8. Relative position of car and two-wheeler at different detection angles before the collision. (a) 2 s before the collision. (b) 1.5 s before the collision. (c) 1 s before the collision.

the two-wheeler being detected at different detection distances were obtained, as shown in Fig. 11.

As shown in Fig. 11, the probability of two-wheeler in fatal accidents target detection decreases with the decrease in the braking advance time [41]. It is also similar for accidents involving injuries at 20 m or more distances. When $t = 2$,

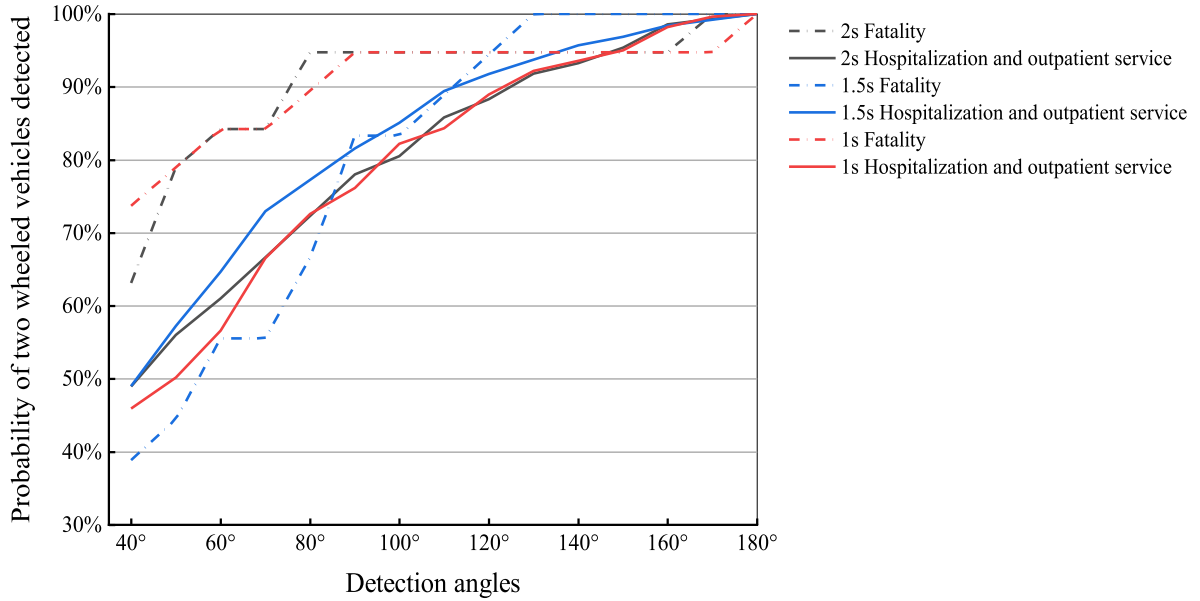


Fig. 9. Probability of two-wheelers being detected at different detection angles at $T = 1, 1.5$, and 2 s.

TABLE II
MILLIMETER WAVE RADAR PRODUCT PARAMETERS

Company	Product Model	Detection Distance/m	Detection Angle/°
Bosch	MRR	160	42
	LRR4	250	30
Continental AG	ARS408	20	120
	ARS410	170	8
	ARS430	250	120
Delphi	ESR2.5	174	20
		60	90

1.5, and 1 s, the two-wheelers were detected at 60, 45, and 35 m, respectively. However, the detection distance depends not only on the relative position of the car and the two-wheeler before the collision, but also on the braking advance time of AEB system. When $TTC = t$, the AEB system will intervene to prevent collisions. The maximum reasonable range of the braking advance time is 0.6–1 s, as determined by the traffic safety laboratory of the United Kingdom [42]. Therefore, when $t = 1$ s, it is enough to detect the potential obstacles. Thus, to ensure the effectiveness of the AEB system in protecting vulnerable road users, the final detection distance should be at least 35 m.

D. Performance Verification of the AEB System

To verify the effectiveness of the optimized AEB system in protecting two-wheeler riders under the current technical conditions, we used the accident cases of 301 screened before. Then, we assumed that all cars in these cases are equipped with the optimized AEB system. The parameters of the optimized AEB system were selected based on the above analysis: a braking deceleration of 6 m/s^2 , braking advance time of 1 s, detection angle of 120° , and detection distance of 35 m. Then, we used the same reconstruction method and PC-Crash software to

reconstruct 301 accident cases again, where cars were equipped with the optimized AEB system. Consequently, we obtained 301 cases of virtual accident results.

Fig. 12 shows an accident case where a car collided with a two-wheeler that was crossing the road from the left side of the intersection. In the virtual simulation, the car stopped at the original collision position, and the two-wheeler stopped after passing the collision position; thus, avoiding the accident. A new vehicle crash speed, V^* , was obtained after all accidents were simulated using the modified parameters for the AEB system. The vehicle crash speed, V^* , of 67 accidents was 0, accounting for 22.3% of the accidents. It means that 22.3% of the accidents could have been avoided using the modified AEB system. Among the remaining 234 accidents, the collision speed of 202 accidents decreased significantly. Meanwhile, that of the other 32 accidents did not decrease significantly due to the failure to detect the presence of obstacles in time. Fig. 13 shows the cumulative distribution of the new and original collision velocities for the remaining 234 accidents.

Under this set of parameters, compared with non-AEB-using cases, the average collision speed of cars decreased from 33.93 to 21.34 km/h, i.e., by 37%. As shown in Fig. 12, the collision speed of the cars did not exceed 48 km/h in 80% of the simulated cases. A previous study [43] shows that the greater the collision speed is, the greater the injury degree of the two-wheeler rider. Therefore, reducing the collision speed can effectively reduce the injury degree of two-wheeler rider.

IV. DISCUSSION

AEB system is an active safety technology. The practical application value of the AEB system testing must be improved through a continuous accumulation of real road traffic accident data. However, many studies have considered traffic accident

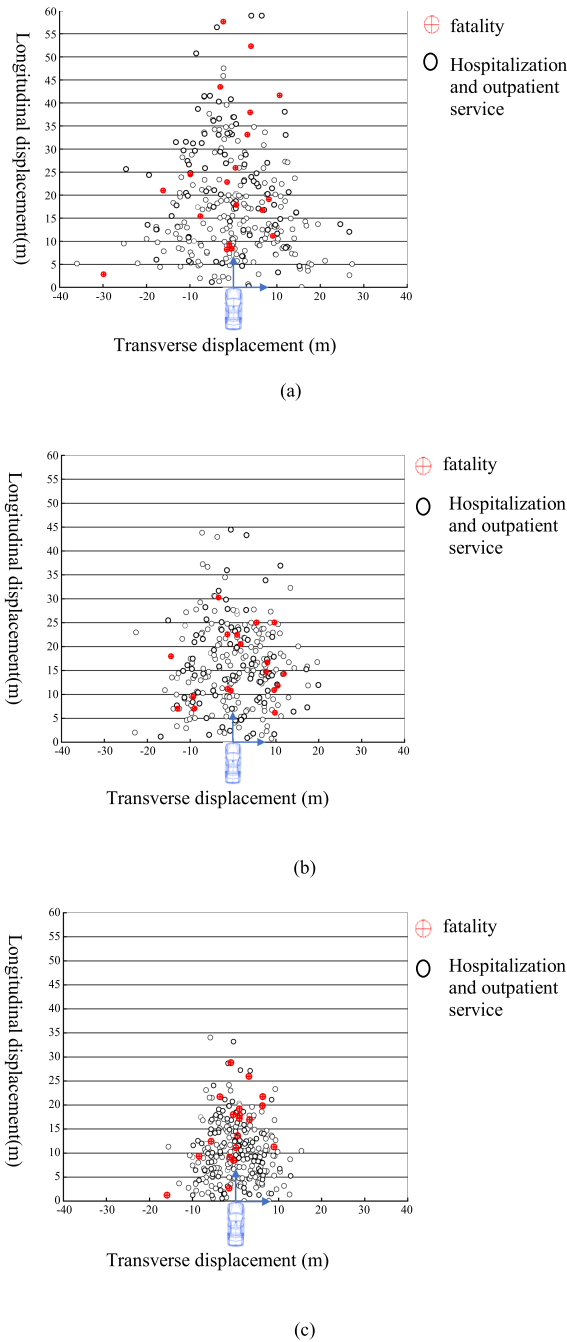


Fig. 10. Relative positions of the car and two-wheeler under different detection distances before collision. (a) 2 s before the collision. (b) 1.5 s before the collision. (c) 1 s before the collision.

cases in a certain area as their research object, which is not representative of the traffic accident situation on a national level. In this paper, we obtain research data from accident cases recorded in the CIDAS. The CIDAS database contains cases from the eight regions of Weihai, Ningbo, Changchun, Beijing, Foshan, Siping, Chengdu, and the southwest of Guizhou Province. These areas include plains, hills, cities, and villages, and the cases range from north to south. The accident data were obtained through on-site investigation by the CIDAS project team's professional accident investigators. The accident investigation team recorded

measurements of the accident scene, drew sketches, and collected specific information about the accident environment and vehicles. Using the victims' injury information ensures that the overall accident data are not affected by the region and ensures the accuracy of the accident site assessment. By using CIDAS data, the research object of this paper ensures the accuracy of the accident data and their representativeness of traffic accidents in the entire country.

C-NCAP, a vehicle safety rating provider, introduced AEB tests for pedestrians. It is expected to enhance the protection of vulnerable road users by including two-wheeler tests in the future. In domestic car to pedestrian collision accidents, when the detection angle of AEB system is set to 50° , and when conditions that may result in severe injuries and deaths can be detected, 92.2% of minor injuries can be prevented. When the detection angle of the AEB system is set to 40° , 96.9% of severe injuries and deaths and 88.3% of minor injuries can be prevented [31]. This study is consistent with the data sources of Liu et al [31]. However, the speed of the two-wheeler is faster than that of pedestrian. Therefore, the lateral distribution of the two-wheelers is wider than that of pedestrians in the same relative position map. Thus, for the same detection angle, the probability of detecting pedestrians is higher than that of two-wheelers.

The results of this study show that the sensor requires 180° coverage to detect all two-wheelers 1–2 s before a collision. Lenard et al. [20] showed that the sensor needed 180° coverage to detect most of the two-wheelers 2–3 s before the collision. The results of the two studies are the same, and part of the difference may be due to the difference in the dangerous traffic conditions between China and Britain.

In this paper, the samples were screened before selection. Accident cases involving a car to two-wheeler collision at the rear of the car or occlusion of the two-wheeler by obstacles were not included. In these accidents, the AEB system would not detect the two-wheeler before a collision. In future research, we will consider developing an AEB system for side-impact avoidance using multiple sensors for the detection or combining it with intelligent vehicle networks [44], [45] to detect the surrounding situation of other vehicles. The detection range of LiDAR technology can reach 360° . However, it has poor penetration ability under extreme weather conditions, a shortcoming that can be compensated using millimeter-wave radars. Therefore, in future research, we will also consider the data fusion of millimeter-wave radars and LiDAR to support each other and expand the detection range of the AEB system to improve its performance.

Based on the existing sensor technology, this paper theoretically analyzed the detection range of the AEB system and verified the performance of the AEB system from the detection range. Consequently, we obtained that 22.3% of the car to two-wheeler collision accidents recorded in China's accident database could have been avoided through using an AEB system. In the unavoidable accidents, the collision speed would have been reduced by 37%, and the number of injuries would have also been smaller. This conclusion is almost consistent with the result of an E-NCAP survey [46].

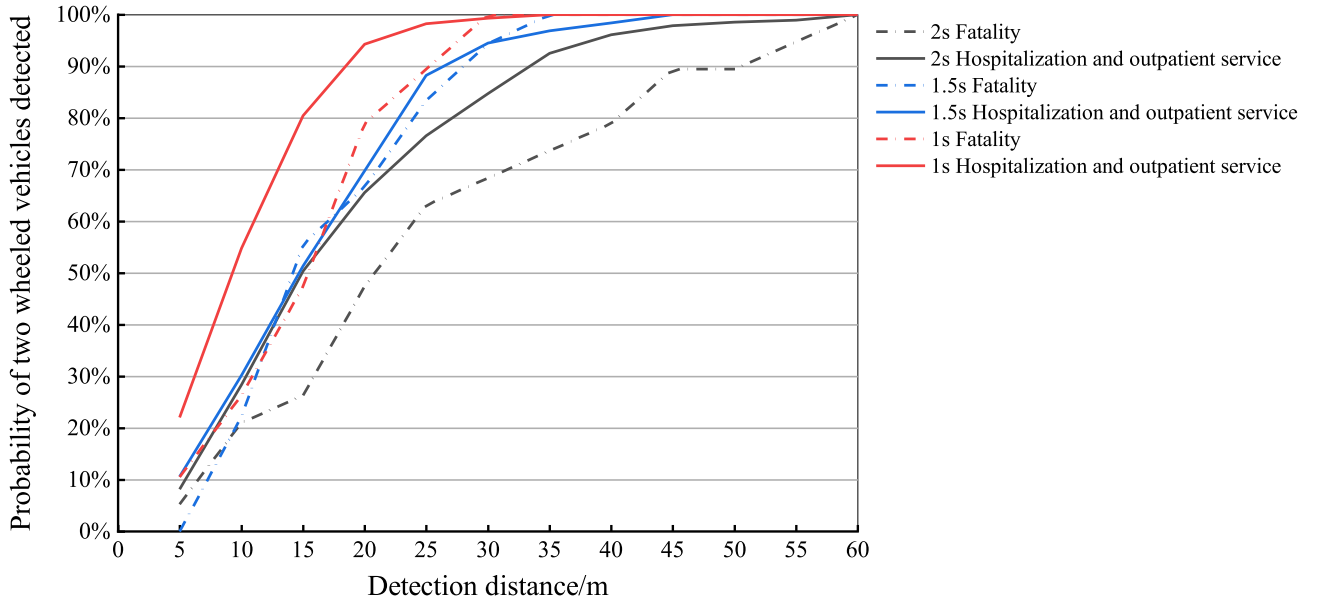


Fig. 11. Probability of two-wheelers being detected at different detection distances when $T = 1, 1.5, 2$ s.

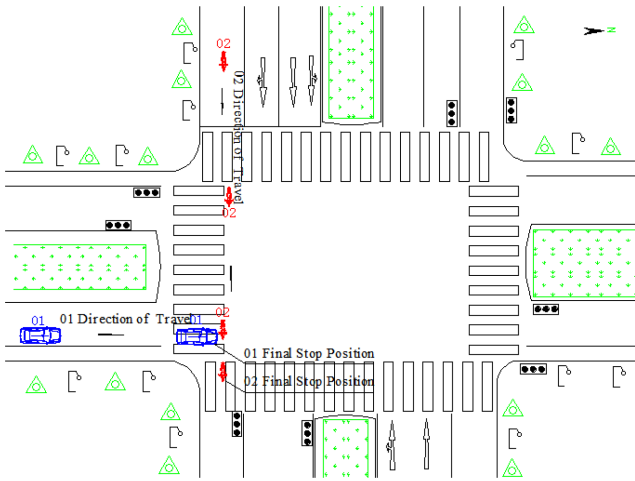


Fig. 12. Virtual accident results after simulation.

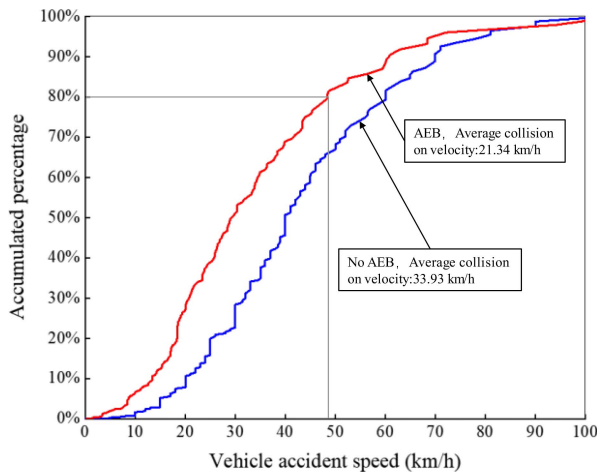


Fig. 13. Accumulated velocity distribution of vehicle accidents with/without AEB system.

V. LIMITATION

The two-wheelers discussed in this paper include bicycles, electric two-wheelers, and motorcycles; however, different types of two-wheelers are not discussed separately. AEB system may have different detection range requirements for different types of two-wheelers.

The results of this study are optimistic. Meanwhile, many two-wheelers cannot be detected by AEB system before the collision because the braking deceleration is set to an ideal value of 6 m/s^2 , and the domestic traffic situation is different from that of foreign countries. This has a certain impact on the detection rate under different detection angles and ranges. In actual road traffic, the braking deceleration is not a constant value, affecting the road conditions. However, in this study, the braking deceleration was set based on the ideal conditions of a dry surface and high-efficiency braking system. Based on this, in future research, the real-time braking deceleration value can be considered, and the braking advance time required for each accident case can be investigated.

According to this study results, this AEB system can avoid 22.3% of car to two-wheeler collision accidents and reduce the average collision speed by 37%, effectively reducing the injury degree of two-wheeler riders. However, in the current research, more studies are needed to prove the system's feasibility.

VI. CONCLUSION

In this paper, the key parameters are obtained through theoretical analysis of AEB system, including braking advance time, braking deceleration, and detection range. According to the theoretical analysis of the AEB braking process, braking advance time mainly consists of brake coordination time and continuous braking time. Among them, we set the brake coordination time to 0.3 s, according to Yang et al. [30]. The continuous braking time is determined by the braking speed

and deceleration. The braking deceleration is affected by the road conditions and cannot be determined directly. The maximum range of the braking deceleration that can be achieved is 5.4–6.9 m/s² [37]. Considering the limitations on the safety and comfort of passengers and common realistic conditions, we used a braking deceleration value of 6 m/s². We selected 301 cases of car to two-wheeler collision accidents in the CIDAS database, which were reconstructed by PC-Crash software. The drivers' behavior and relative positions of car and two-wheeler within 2 s before the collision was determined based on the accident reconstruction. The results show that the vehicle driver had not taken any measures within 2 s before the collision in 53% of these 301 cases. To ensure the protective effect of the AEB system for two-wheeler riders in the car to two-wheeler collision accidents, the detection range of the AEB system was optimized combined with braking advance time and braking deceleration. The optimized ranges are summarized as follows: (1) Theoretical results gave a detection angle and distance of 180° and 35 m, respectively. (2) Technically feasible results produced a detection angle and distance of 120° and 35 m, respectively. We applied these optimized parameters of the AEB system to 301 accident cases and reconstructed them to verify the performance of the optimized AEB system. The main results are as follows: (1) The optimized AEB system can avoid 22.3% of the car to two-wheeler collision accidents. (2) When the detection angle is 120°, and the detection distance is 35 m, in accidents where collisions still occur, the average crash speed decreases from 33.93 km/h for cars without an AEB system to 21.34 km/h for cars with an AEB system, i.e., a reduction of 37%. The experimental results show that the optimized AEB system can effectively reduce the occurrence of car to two-wheeler collision accidents and the severity of unavoidable accidents.

In terms of promoting application value, findings of this study can help car manufacturers and makers of AEB hardware to set up an AEB product development strategy. It can also provide data support for building/improving relevant laws and regulations about the AEB system in China, such as AEB testing and evaluation for vulnerable road users in C-NCAP and C-IASI. In terms of academic research, this study can provide ideas and fundamental knowledge for the future development of AEB system or other driver assistance systems.

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